

Week 2 – Spatial Description and Transformation

ELEC0129 Introduction to Robotics

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Schedule

Legend:

A: MPEB 6.02 Teaching Lab

B: Roberts 7.04 PC Lab

Week	Recorded, uploaded by Friday of previous week	Workshop, Mondays 11am-1pm and/or Wednesdays 9am-11am	Notes
1	(Scenario Week)		
2	Lec: Intro; Spatial description	A: Workshop: Offline programming	Half class Mon, half class Wed
3	Lec & Tut: Spatial description	A: Workshop: Build robot	
4	Lec & Tut: Forward kinematics	A: Workshop: Forward kinematics	
5	Lec & Tut: Inverse kinematics	B: Workshop: Offline programming	Half class Mon, half class Wed
RW	(Reading Week)		
6	(Scenario Week)		
7	Lec & Tut: Jacobians	A: Workshop: Inverse kinematics	
8	Lec: Trajectory Planning	A: Workshop: Trajectory planning	
9	Lec & Tut: Dynamics	A: Workshop: Trajectory planning	
10	Lec & Tut: Control	A: Workshop: Pick-and-place demo	

Content (What you will learn today)

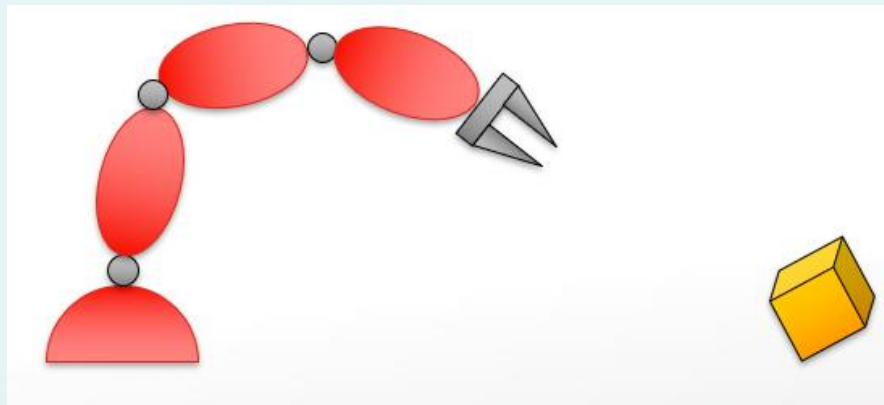
- Description of position and orientation
- Mapping
- Transformation Arithmetic

Content

- Description of position and orientation
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- Transformation Arithmetic

Introduction: Description

- Robotic manipulation → Parts and tools moved around in space.
- Need to know how to **describe** position and orientations of objects.



Position of a Point (1)

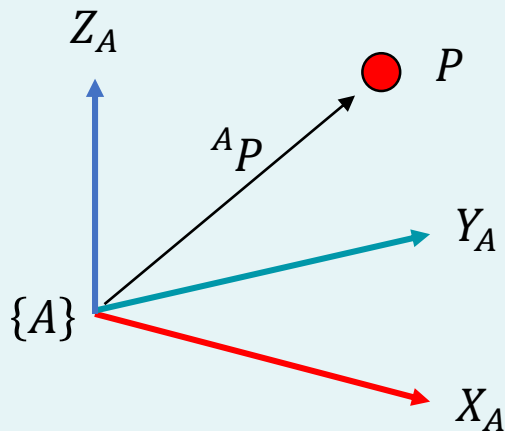
- How do you describe the position of the point P below?



- You will need a reference frame (coordinate).
- E.g.
 - 15cm from the left edge of this screen,
 - 10cm from the bottom edge of this screen,
 - 0cm out of plane of this screen.
- Or:
 - 5m from the left wall of this classroom,
 - 2m from the front wall of this classroom,
 - 1.5m above the floor of this classroom.

Position of a Point (2)

- Any point can be described with respect to any coordinate system using a 3 x 1 position vector.
- Notation:

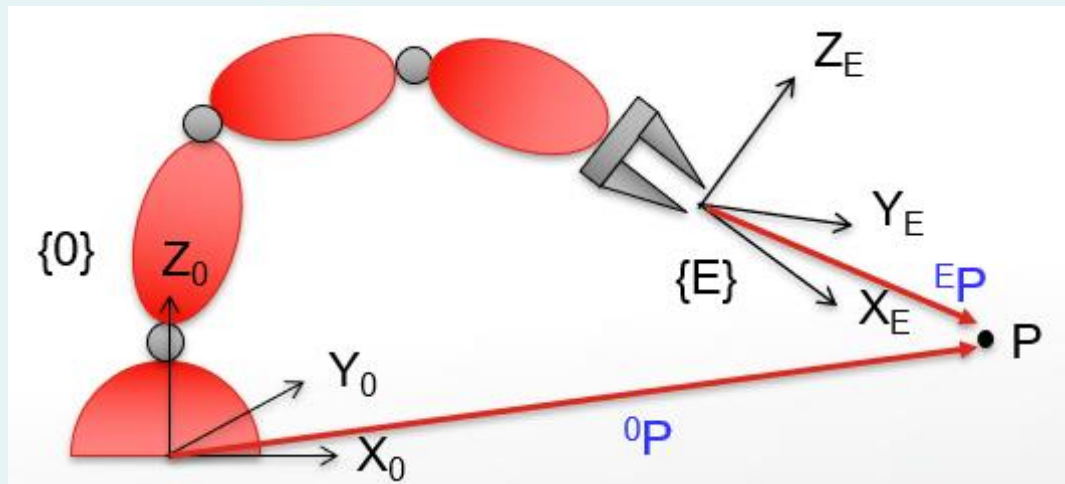


- In frame $\{A\}$, the point P has a position vector

$${}^A P = \begin{bmatrix} {}^A P_x \\ {}^A P_y \\ {}^A P_z \end{bmatrix}.$$

Position of a Point (3)

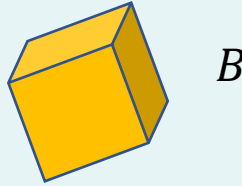
- Examples:



- Position of the point P with respect to robot base frame {0} is 0P .
- Position of the same point with respect to the end-effector frame {E} is EP .
- Note: They are the same point, just in different frames!

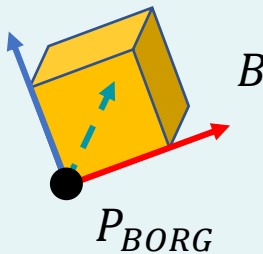
Position of a Rigid Body (1)

- How do we describe the position of a rigid body B ?



Position of a Rigid Body (2)

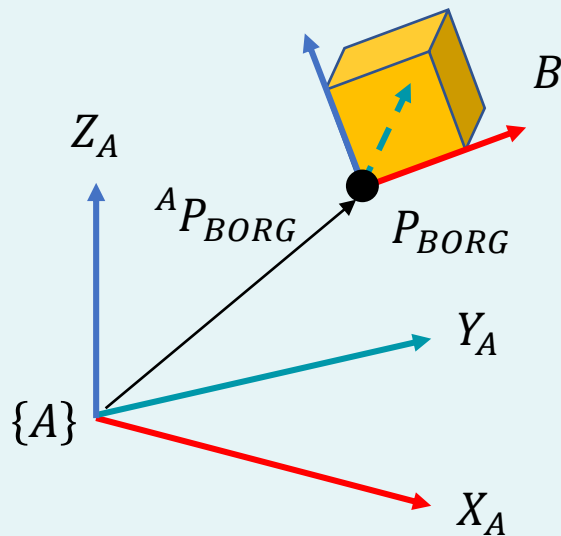
- Attach a **frame** to the object.



- Use the **origin of the frame** as the point for position description.

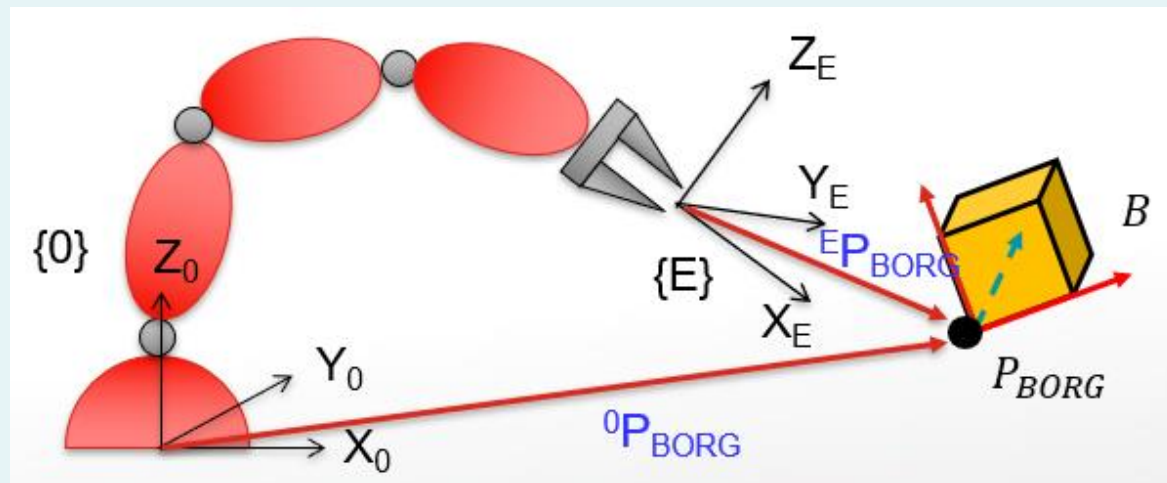
Position of a Rigid Body (3)

- You will still need the **reference frame**!
- E.g. in frame $\{A\}$, the object B has a position vector ${}^A P_{BORG}$.



Position of a Point (4)

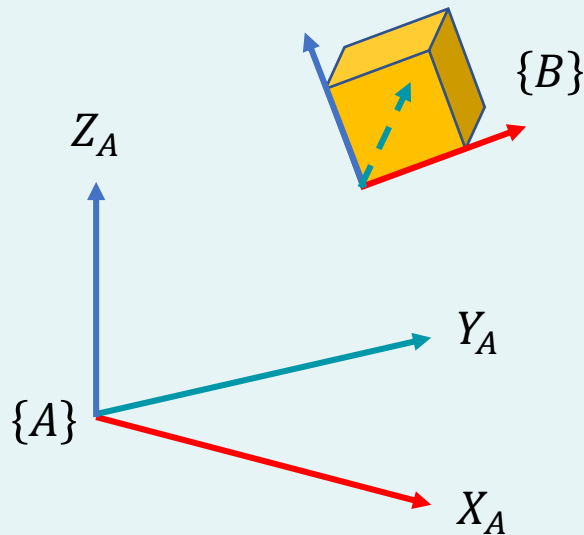
- Examples:



- Position of object B with respect to robot base frame {0} is ${}^0P_{BORG}$.
- Position of the same object with respect to the end-effector frame {E} is ${}^EP_{BORG}$.

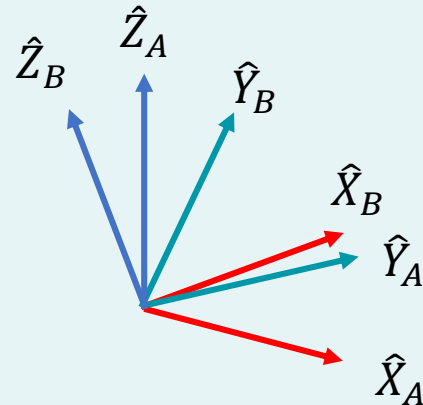
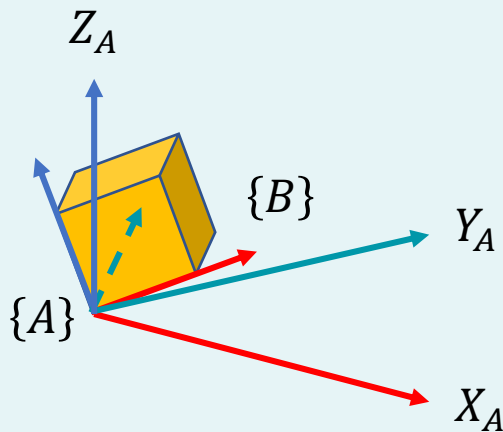
Orientation of a Rigid Body (1)

- What about **orientation**?
 - Use the same frames for the rigid body as in earlier slide.



Orientation of a Rigid Body (2)

- Then ignore the translation and let the **origins of frames coincide**.
- Next, scale the axes such that they become unit vectors.

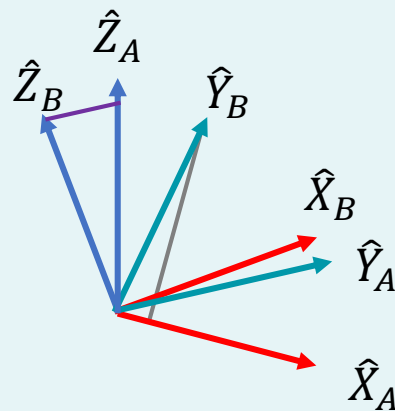


Orientation of a Rigid Body (3)

- The **orientation** of a rigid body {B} with respect to reference frame {A} is now defined as a **rotation matrix**: Projection of $\hat{Y}_B$ onto $\hat{X}_A$.

$${}^A_B R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

Projection of $\hat{Z}_B$ onto $\hat{Z}_A$.



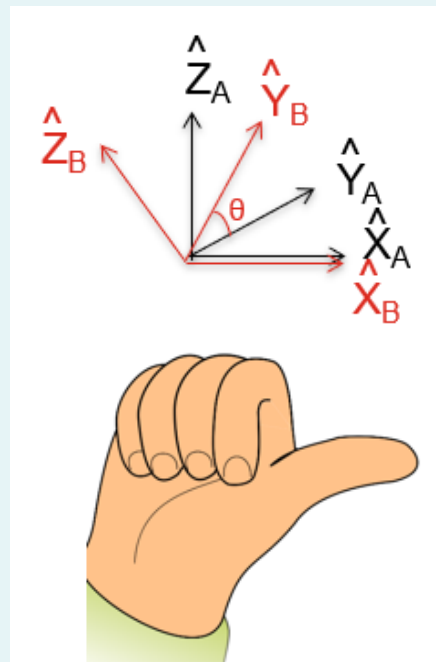
- where $\hat{X}, \hat{Y}, \hat{Z}$ means the **unit vector** of X, Y, Z respectively.

Orientation of a Rigid Body (4)

- E.g. rotation about x-axis only:

$${}^A_B R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

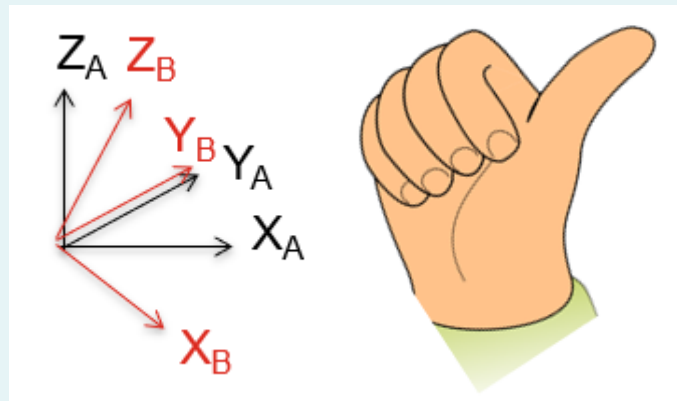
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$



Orientation of a Rigid Body (5)

- E.g. rotation about y-axis only:

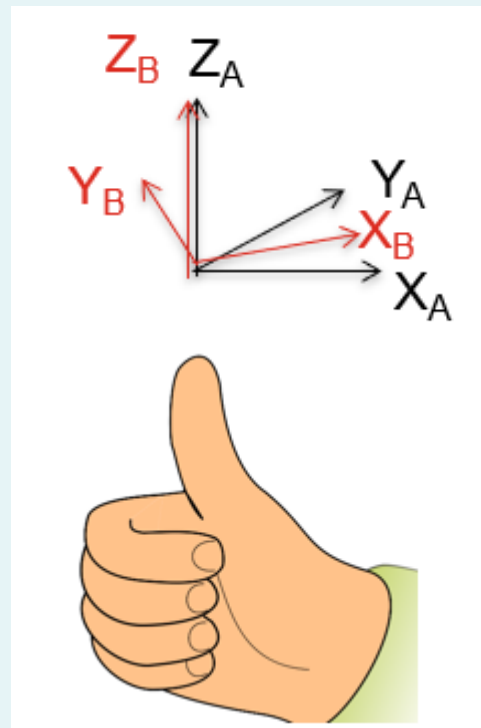
$$\begin{aligned} {}^A_B R &= \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \end{aligned}$$



Orientation of a Rigid Body (6)

- E.g. rotation about z-axis only:

$${}^A_B R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$
$$= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Orientation of a Rigid Body (7)

- The rotation matrix is **orthonormal**.
 - Each column and each row has **unit magnitude**; and
 - Each column is **orthogonal** (90 degrees) to other columns, i.e. the dot products between two columns are zero.

- E.g. ${}^A_B R = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- Therefore, the **inverse** is the same as **transpose**.

- ${}^B_A R = {}^A_B R^{-1} = {}^A_B R^T$

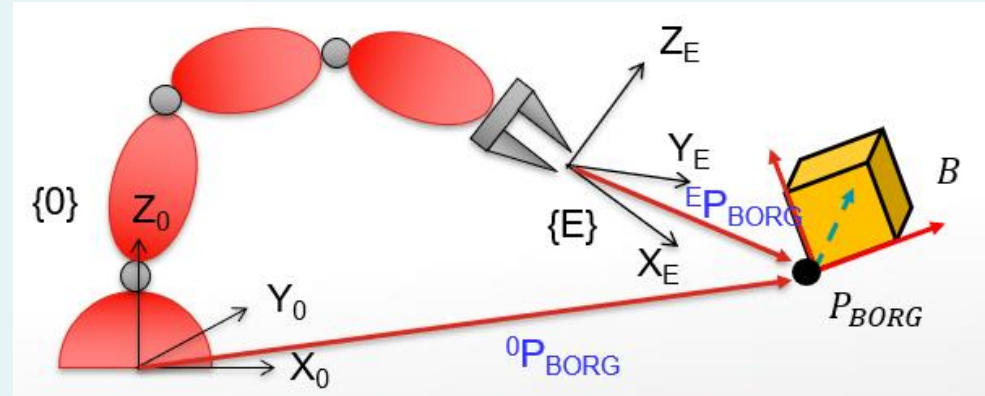
Description of a Frame

- We have looked at position and orientation separately.
- To completely specify the whereabouts of the tool / object etc., we need both the position and orientation.
- Use the **rotation matrix + the position vector of frame origin** to specify:

- $\{B\} = \{ {}^A_B R, {}^A P_{Borg} \}$ e.g.

- $\{B\} = \{ {}^0_B R, {}^0 P_{Borg} \}$

- $\{B\} = \{ {}^E_B R, {}^E P_{Borg} \}$

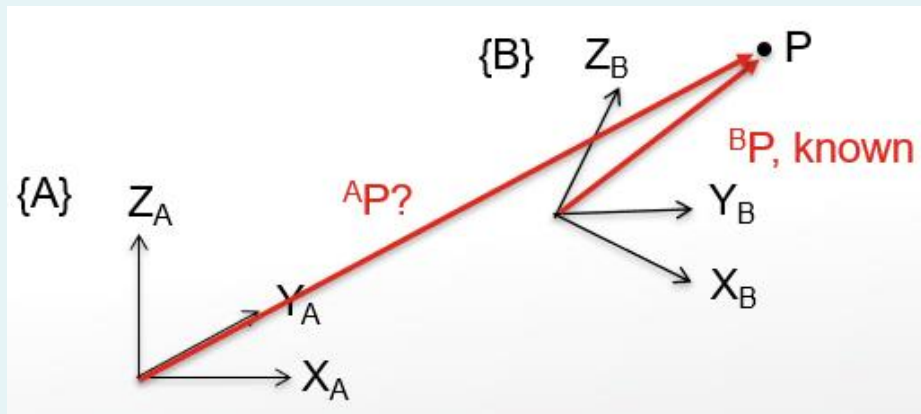


Content

- Description of position and orientation
- Mapping
- Transformation Arithmetic

Introduction: Mapping

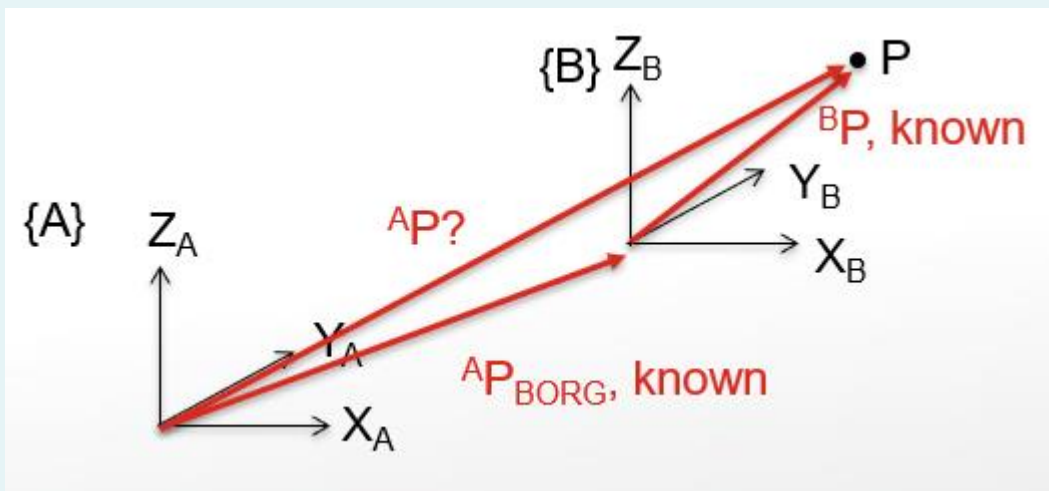
- In mapping, the question is:
 - Given a point P, which is already described / known in coordinate system {B}.
 - What is the position of P in coordinate system {A}?
- Important: The point itself is not changed! Only its description is changed.



Mapping involving Translated Frames

- If the frames differ only by a pure translation, then:

- $${}^A P = {}^B P + {}^A P_{Borg}$$

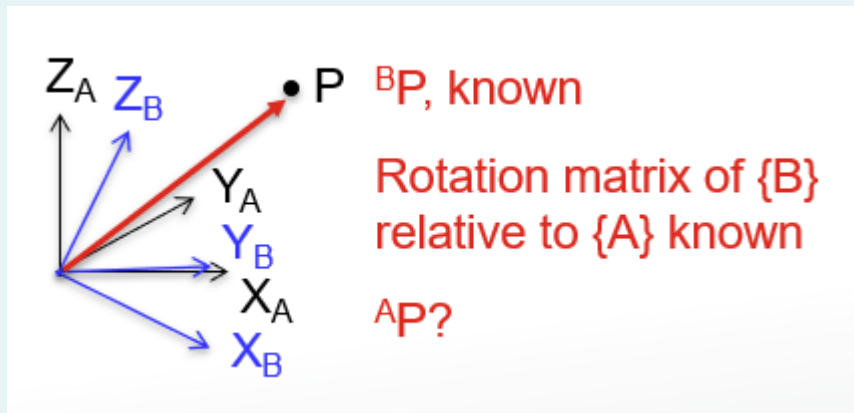


- Caution: Vector addition is allowed **only if the orientation of the frames are the same!**

Mapping involving Rotated Frames (1)

- If the frames differ only by a pure rotation, then:

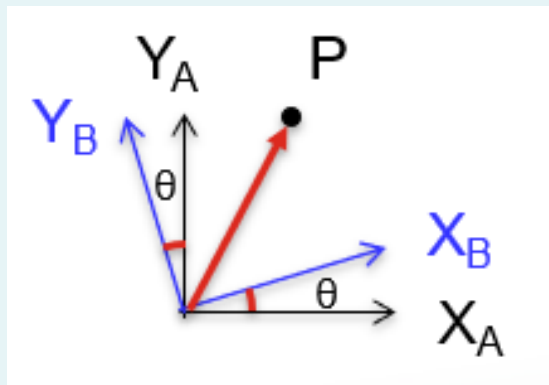
- $${}^A P = {}^A_B R \cdot {}^B P$$



- Note: The notation is useful to keep track of mappings and frames of reference. E.g. cancellation of both B's in the above equation.

Mapping involving Rotated Frames (2)

- Example:
 - Frames $\{B\}$ is obtained by rotating $\{A\}$ along the z-axis by 30deg.
 - In Frame $\{B\}$, the point P has a coordinate of $[1, 1, 0]^T$.
 - What is the coordinate of point P in frame $\{A\}$?



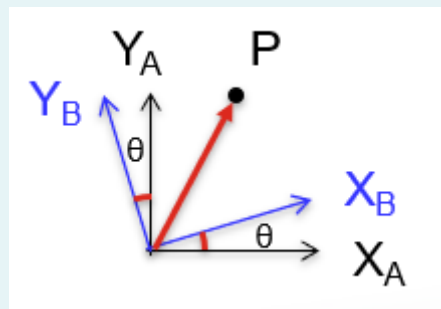
Mapping involving Rotated Frames (3)

- Solution:
 - First, get description of {B} in {A}.

$${}^A_B R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

$$= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

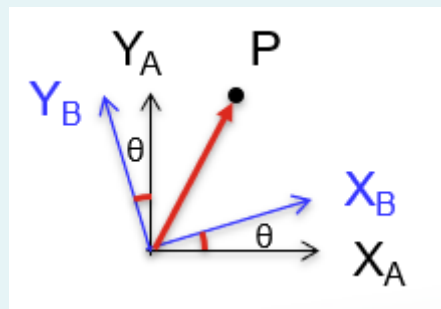
$$= \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Mapping involving Rotated Frames (4)

- Secondly, calculate:

$$\begin{aligned} \bullet \quad {}^A P &= {}^A_B R \cdot {}^B P \\ &= \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0.366 \\ 1.366 \\ 0 \end{bmatrix} \end{aligned}$$

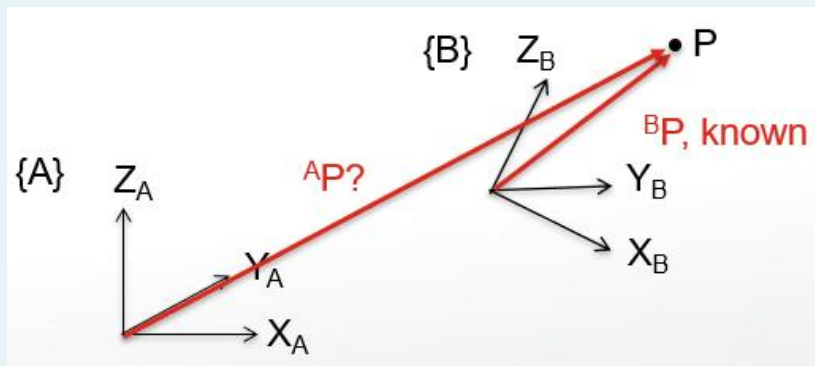


- The answer looks right from the figure.

Mapping involving both T and R (1)

- If the frames now differ by both translation and rotation:
 - First, change the description of ${}^B P$ into a frame which has the **same orientation** of $\{A\}$.
 - Then perform **vector addition** for translation. (Remember? Vector addition is allowed only if the frames have same orientation).
- The formula is:

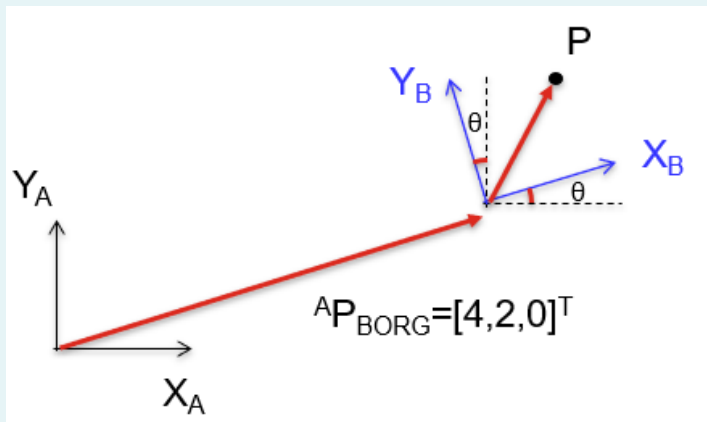
$${}^A P = {}^A_B R \cdot {}^B P + {}^A P_{Borg}$$



Mapping involving both T and R (2)

- Example: Same rotation as in previous example, but now with translation of the frames:

$$\bullet \quad {}^A P = {}^A_B R \cdot {}^B P + {}^A P_{Borg} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 4.366 \\ 3.366 \\ 0 \end{bmatrix}$$



Mapping involving both T and R (3)

- The equation ${}^A P = {}^A_B R \cdot {}^B P + {}^A P_{Borg}$ is not very appealing.
- Can we get the same answer with just a simple matrix multiplication, i.e. ${}^A P = {}^A_B T \cdot {}^B P$?
- Yes! By a clever trick:

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B R & {}^A P_{Borg} \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

- The 4 x 4 matrix operator is called “Homogeneous Transform”.

Mapping involving both T and R (4)

- Same example as just now:

$$\begin{bmatrix} A_P \\ \hline 1 \end{bmatrix} = \begin{bmatrix} & A_B R & & \\ \hline 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} B_P \\ \hline 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 & 4 \\ 0.5 & 0.866 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4.366 \\ 3.366 \\ 0 \\ 1 \end{bmatrix}$$

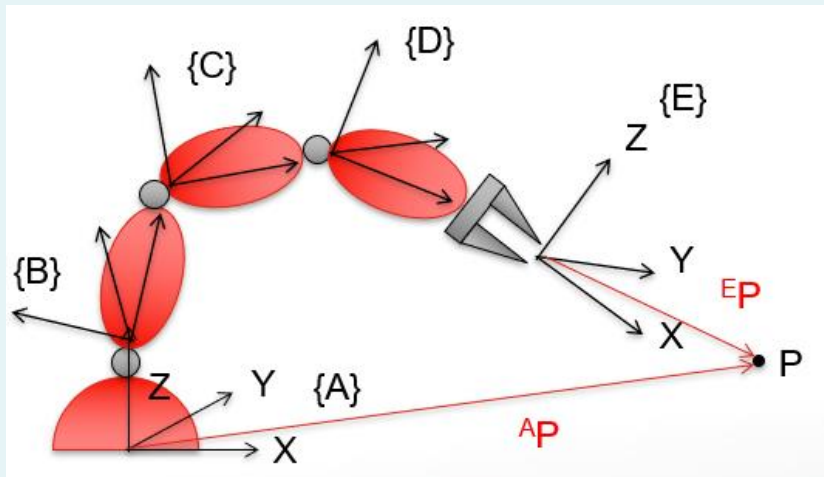
- Just ignore the last row from the final result.

Content

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Compound Transformation (1)

- Robots usually have a few joints and links.
 - The coordinate of P in frame {E}, i.e. ${}^E\mathbf{P}$ is known.
 - {E} relative to {D} is known; {D} relative to {C} is known; {C} relative to {B} is known, and finally {B} relative to {A} is known.
 - What is ${}^A\mathbf{P}$?

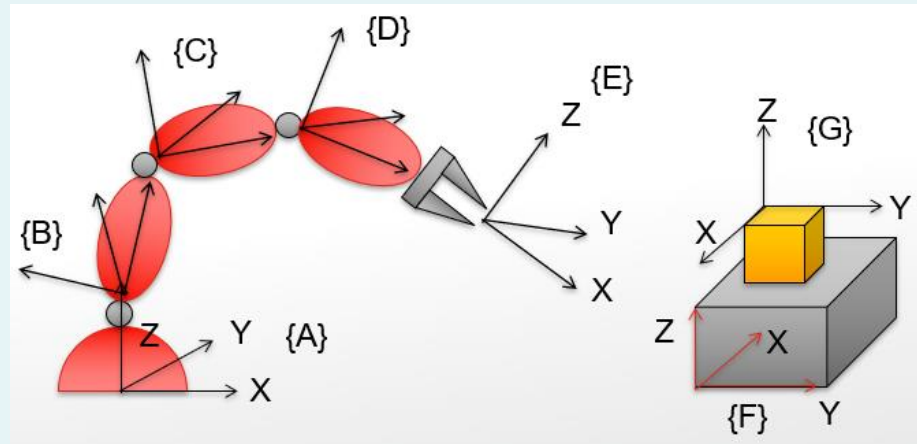


Compound Transformation (2)

- The process is straightforward: Do “mapping” step by step.
 - ${}^E P$ given.
 - ${}^D P = {}^D_E T \cdot {}^E P$.
 - ${}^C P = {}^C_D T \cdot {}^D P = {}^C_D T \cdot {}^D_E T \cdot {}^E P$.
 - ${}^B P = {}^B_C T \cdot {}^C P = {}^B_C T \cdot {}^C_D T \cdot {}^D_E T \cdot {}^E P$.
 - ${}^A P = {}^A_B T \cdot {}^B P = {}^A_B T \cdot {}^B_C T \cdot {}^C_D T \cdot {}^D_E T \cdot {}^E P$.
- Note: The notation is useful to keep track of mappings and frames of reference. E.g. cancellation of B's, C's etc. in the last equation.

Transform Equation (1)

- In the following figure, assume you know the following:
 - All transformations from robot base to end-effector: A_BT , B_CT , C_DT , D_ET
 - Transformation from robot base to table, and table to object: A_FT , F_GT



- Then what is the transform from **end-effector to object** E_GT ?

Transform Equation (2)

- It is clear that: $\boxed{{}_G^AT = {}_B^AT \cdot {}_C^BT \cdot {}_D^CT \cdot {}_E^DT \cdot {}_G^ET = {}_F^AT \cdot {}_G^FT.}$

- Solve for the unknown ${}_G^ET$:

- ${}_C^BT \cdot {}_D^CT \cdot {}_E^DT \cdot {}_G^ET = {}_B^AT^{-1} \cdot {}_F^AT \cdot {}_G^FT.$

- ${}_D^CT \cdot {}_E^DT \cdot {}_G^ET = {}_C^BT^{-1} \cdot {}_B^AT^{-1} \cdot {}_F^AT \cdot {}_G^FT.$

- ${}_E^DT \cdot {}_G^ET = {}_D^CT^{-1} \cdot {}_C^BT^{-1} \cdot {}_B^AT^{-1} \cdot {}_F^AT \cdot {}_G^FT.$

- $\boxed{{}_G^ET = {}_E^DT^{-1} \cdot {}_D^CT^{-1} \cdot {}_C^BT^{-1} \cdot {}_B^AT^{-1} \cdot {}_F^AT \cdot {}_G^FT.}$

- You will learn how to calculate the **inverse** (of 4x4 matrix) easily next week.

Thank you for your attention!

Any questions?